Build Overview:

1 **Ground Mecca Install**
2 **Block Off Panels Install**
3 **SMC Tray Install**
4 **Ethernet Switch Install**
5 **Operator Side EMO Switch Install**
6 **Membrane Panels Install**
7 **Operator Interface Panels Install**

CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by	
	1	New Issue	JOSEPH AMAD	**Prepared By:** JOSEPH R. AMAD, Manufacturing Engineer
				Checked By: IVAN NOWELL PANIQUE, Team Lead
				Approved By: SHERWIN ARIOSTE, Operations Manager

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	633-799-00	ELEC-MECH, UF2 24SLOT SUPPORT CABINET INTEGRATION	1

MACHINE: N/A

METHOD: 1. Scope of application

This BPM is suitable for UF Plus Support Cabinet Assembly
Applicable Support Cabinet Model 633-799-00
The sub-part model in Support Cabinet are 629-502-01 ADB CHASIS SUB ASSY, 653-315-00 CDBU SUB ASSY and 630-945-10 GEMINI SMC TRAY

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	633-276-00	FORGED OR STAMPED STEEL ME GROUNDING BLOCK	1
2	614-766-90	FASTNR,RIP-017-1, 12X30X10MM HT. M8 THRD. HOLE, ZN.	2
3	470-933-01	SCREW CAP SKT BUTHD M8X1.25 20MML SS	2
4	470-501-06	WASHER, FLAT, 8.4MM ID, 16MM OD, SS	2
5	470-502-07	WASHER LK SPLT M8, 14.8MM OD, SS	2
6	632-132-00	FORGED OR STAMPED STEEL, BLANK OFF, PASS-THRU	4
7	603-096-00	NUT, HEX, SS M8X1.25, WSHR W/TTH, 13MM WDH	41
8	938-172-01	WI TYWRAP ANCHOR #10SCREWMTG	2
9	631-823-00	CDU "Z" Bracket	1
10	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 12MM L	9
11	653-315-00	CDU	1
12	630-945-10	ELEC-MECH, UF2 SMC TRAY, FOR SEMI TEST	1
13	472-426-20	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 20MM L	2
14	666-796-00	CATALOG, NETGEAR 10/100MBPS FAST ETHERNET SWITCH	1
15	479-133-23	LABEL, EMERGENCY OFF	1
16	801-513-08	SW RED PUSHBUTN OPERATOR GUARD	1
17	801-643-01	HDWR SW OPERATOR TURN/RELEASE	1
18	801-541-02	SW CONTACT BLOCK,NO	1
19	801-515-02	SW CONTACT BLOCK 1F	1
20	634-438-00	CABLE EMO BUTTON CABLE W/CONN	1
21	631-021-00	CABLE MEMBRANE SYSTEM STATUS PNL LESS THAN 80V V	1
22	631-018-00	FORGED OR STAMPED STEEL BACKING PLATE SYSTEM	2
23	631-020-00	CABLE MEMBRANE SYS TEST PWR SW LESS THAN 80V V	1
24	901-849-01	LC ADAPTER (EMI) 45DEG SCRW MT	1
25	630-893-00	FORGED OR STAMPED STEEL CAB I/O PANEL SIGNAL MISC	1
26	406-501-05	SCRW PAN HEAD TORX M2X.4PX5MM SS W/NYLOK	2
27	357-725-00	PANLMNT CAT5E RJ45 FEEDTHRU	2
28	634-733-00	CABLE CAN I/O PANEL EMO 24 V W/ CONN	1
29	634-736-00	CABLE CAN I/O PANEL EMO 24 V W/ CONN	1
30	633-966-00	FORGED OR STAMPED STEEL CAB I/O PANEL SPARE MISC	1
31	654-193-01	HDW, NUT, HEX CAP M5, 7.9MM W, POLYETHYLENE, BLACK	4

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	5mm Hex Bit	13.5 N.m
2	8mm Socket Head	3.3 N.m
3	8mm Socket Head	1.6 N.m
4	Lifting Hook and Lift Truck	N/A
5	CDU Trolley	N/A
6	PH2 Bit	1.6 N.m
7	PH1 Bit	0.7 N.m
8	EMO Switch Tool	17.7 lbs/ 2.0 N.m
9	Flat-Head Torque Screwdriver	0.7 N.m
10	T6 Torx Bit	0.19 N.m
11	PH2 Bit	3.3 N.m

METHOD:

Flow chart of operation steps

BPM preparation	See (1. Scope of application)
↓	
Preparation Tools	See (2. Tools/Materials/Parameters---1. Tools)
↓	
Prepare Materials	See (2. Tools/Materials/Parameters---2. Materials)
↓	
Complete Operation	See (3. Operation Steps)
↓	
Inventory	See (2. Tools/Materials/Parameters---2. Materials)
↓	
Operation Check	See (4. Matters needing attention)
↓	
Inventory collection tools	See (2. Tools/Materials/Parameters---1. Tools)
↓	
Inventory collections tools	See (2. Tools/Materials/Parameters---1. Tools)

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	632-276-00	FORGED OR STAMPED STEEL ME GROUNDING BLOCK	1
2	614-766-90	FASTNR, RP-017-1, 12X30X10MM HT. M8 THRD. HOLE, ZN.	2
3	470-993-01	SCREW CAP SKT BUTHD M8X1.25 20 MML SS	2
4	470-501-06	WASHER, FLAT, 8.4 ID, 16MM OD, SS	2
5	470-502-07	WASHER LK SPLT M8, 14.8MM OD, SS	2

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	5mm Hex Bit	13.5 N.m

METHOD:

1.1 Install a grounded MECCA panel (633-276-00) on the right side of the operation surface of the cabinet as shown in the figure, and fix it with the following materials. The tool is a 5mm hexagon socket head.The torque value is 13.5 N·m:

a. 2 Nut Block (614-766-90)

b. 2 M8 Large Stud Screws (470-933-01)

c. 2 M8 flat washers (470-501-06).

d. 2 M8 spring pads (470-502-07).

Note: The direction of the Large diameter G1 and G2

Note: Please hold the pressure block when installing the screw, otherwise the pressure block will fall.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	632-132-00	FORGED OR STAMPED STEEL, BLANK OFF, PASS-THRU	1
2	603-096-00	NUT, HEX, SS M8X1.25, WSHR W/TTH, 13MM WDH	2

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	5mm Hex Bit	13.5 N.m

METHOD:

2.1 Install 3 baffles (632-132-00) on the operation surface of the cabinet as shown in the figure, and fix them with 12 M5 hexagon nuts (603-096- 00), The tool is an 8mm socket wrench with a torque value of 3.3 N·m.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	632-132-00	FORGED OR STAMPED STEEL, BLANK OFF, PASS-THRU	1
2	603-096-00	NUT, HEX, SS M8X1.25, WSHR W/TTH, 13MM WDH	4
3	938-172-01	WI TYWRAP ANCHOR #10SCREWMTG	2

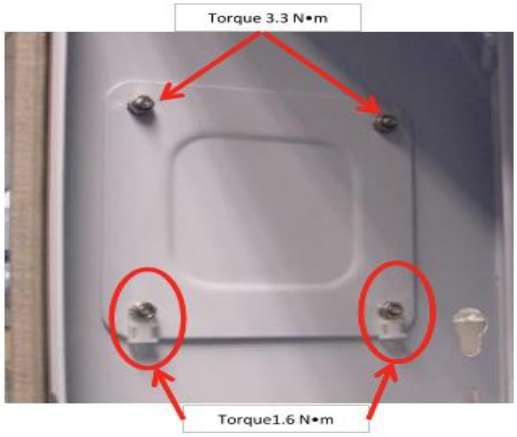
MACHINE:

NO.	DESCRIPTION	PARAMETER
1	8mm Socket Head	3.3 N.m
2	8mm Socket Head	1.6 N.m

METHOD:

2.2 Install a baffle plate (632-132-00) on the operating surface of the cabinet as shown in the figure, and use two M5 hexagon nuts (603-096-00) to fix the two positions above. The tool is an 8mm socket wrench with a torque value of 3.3 N·m.

2.3 As shown in the figure, install two tie buckles (938-172-01) at the two studs at the bottom of the baffle on the upper left side of the cabinet and then use two M5 hexagon nuts (603-096-00) Fix the two positions below. The tool used is an 8mm socket wrench with a torque value of 1.6 N.m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	630-945-10	ELEC MECH, UF2 SMC TRAY, FOR SEMI TEST	1
2	472-426-20	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 20MM L	2

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	PH2 Bit	3.3 N.m

METHOD:

3.1 Take an SMC (630-945-10) as shown in the figure and remove the 2 cable ties shown in the figure.

3.2 Install the SMC from the operating surface of the cabinet.

3.3 Use 2 screws (472-426-20) to fix the SMC bracket on the cabinet. The tool is a PH2 Phillips torque screwdriver with a torque value of 3.3 N·m.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	603-096-00	NUT, HEX, SS M8X1.25, WSHR W/TTH, 13MM WDH	1

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	8mm Socket Head	3.3 N.m

METHOD:

3.4 The view as shown in the figure is a schematic diagram of the SMC of the cabinet operation surface.

3.5 Connect the SMC ground wire through the guide rail to the MECCA G11 interface of the ground panel as shown in the figure, and fix it with a M5 hexagon nut (603-096-00), The tool is an 8mm socket wrench with a torque value of 3.3 N·m.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	603-096-00	NUT, HEX, SS M8X1.25, WSHR W/TTH, 13MM WDH	1

MACHINE: Torque 0.7Nm, Extension Bit, Bit PH2

METHOD: **Preparation of Support Cabinet Network Switch**

1. Take an Network Switch (688-581-00)
2. Install the bracket on the switch as shown in the figure. Use the built-in fixed material to fix the bracket. The tool is a PH2 Phillips torque screwdriver, and the torque value is 0.7 N·m
3. Discard the power cord that comes with the switch.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	472-426-12	M5 Screw	4

MACHINE: Torque 3.3Nm, Bit PH2

METHOD: **Installation Network Switch to Cabinet**

1. In Operation Side view, take Network Switch Assembly and hang it with 4 pcs M5 Screws (472-426-12). The loacation of cabinet rail 36U and 36L both side
2. Push up The Network Switch while torque 4 pcs M5 Screws - Torque to 3.3Nm

MAN: CERTIFIED OPERATOR, TECHNICIAN

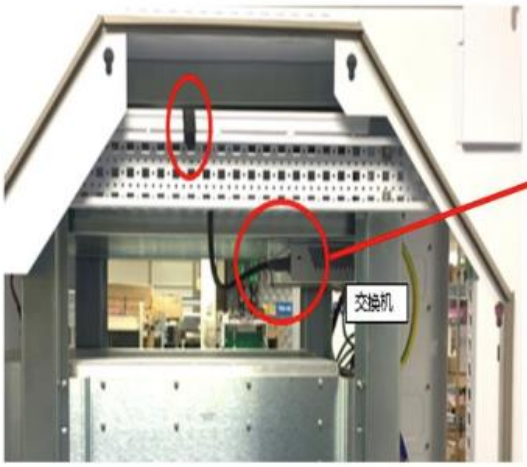
MATERIAL:

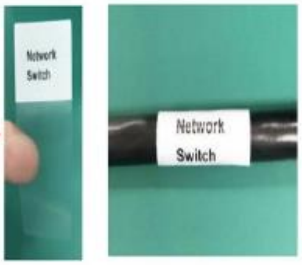
ITEM	P/N	DESCRIPTION	QTY
1	918-241-02	AC Power Cord	1
2	813-929-00	Long Velcro	1

MACHINE: N/A

METHOD: **Installation Network Switch power cord**

1. Obtain 1 pcs AC power cord (918-241-02)
2. Label Ac Power Cord from ADB end with Sticker Label 'ADB J14B' and other with Sticker Label 'Network Switch' as shown in diagram.
3. Plug marked 'Network switch' side of power cord into Network Switch
4. Route to the left side AC cable towards the Maintenance side of the cabinet. Secure the cable with 1 pcs Velcro Tie (813-929-00)





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	479-133-23	LABEL, EMERGENCY OFF	1

MACHINE: N/A

METHOD:

5.1 Install one EMO warning label (479-133-23) on the interface panel of the cabinet operation surface as shown in the figure.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

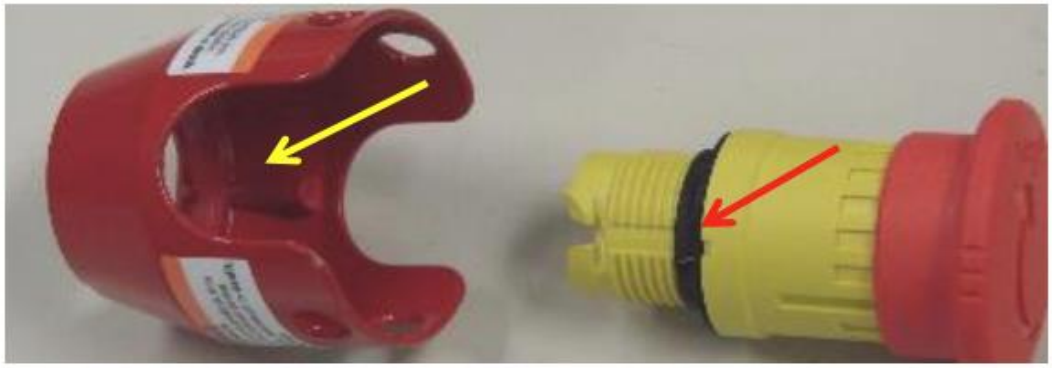
ITEM	P/N	DESCRIPTION	QTY
1	801-513-08	SW RED PUSHBTN OPERATOR GUARD	1
2	801-643-01	HDWR SW OPERATOR TURN/RELEASE	1

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	EMO SWITCH TOOL	17.7lbs/ 2.0 N.m

METHOD:

5.2 Install a shield (801-513-08) to the EMO switch (801-643-01). Use the buckle on the top of the shield as shown in the picture Click on the EMO switch.



5.3 Install the EMO switch (801-643-01) into the hole in the Interface area on the left side of the operation surface of the cabinet. Pay attention to the switch direction when installing the switch, The uppermost text of the switch faces upwards. After installation, use a large plastic nut to lock the switch. The tool is the icon switch torque wrench, the torque value 17.7 in.lbs/ 2.0 N.m

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	801-541-02	SW CONTACT BLOCK, NO	1
2	801-515-02	SW CONTACT BLOCK 1F	1
3	634-438-00	CABLE EMO BUTTON CABLE W/ CONN	1

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	Flat-Head Torque Screwdriver	0.7 N.m

METHOD:

5.4 Install the second connector (801-541-02) to the main connector (801-515-02) as shown in the figure.

5.5 Follow the steps below to install the wire (634-438-00) on the switch interface as shown in the figure.

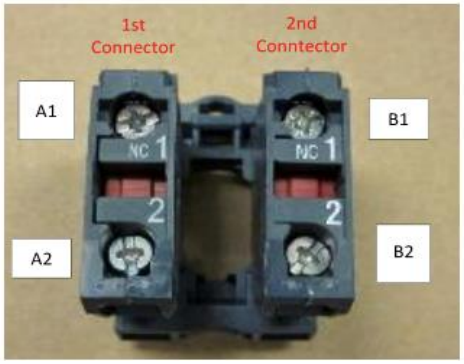
a. A1 is connected to terminal 1 of the 1st connector.

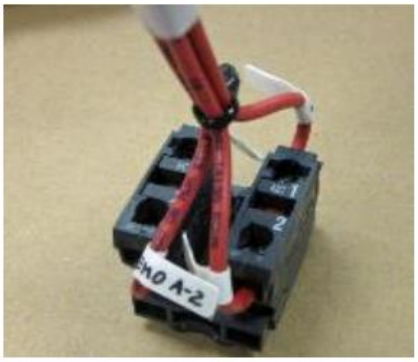
b. B1 is connected to terminal 2 of the 1st connector.

c. A2 is connected to terminal 1 of the 2nd connector.

d. B2 is connected to terminal 2 of the 2nd connector.

Use a flat-head torque screwdriver with a torque value of 0.7 N·m.





5.6 With the switch port facing upwards, install the EMO connector switch to the back position of the EMO button. Gently push the button to ensure that the EMO switch can It is pressed in and rotated out normally without obstacles.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	631-021-00	CABLE MEMBRANE SYSTEM STATUS PNL LESS THAN 80V V	1
2	631-018-00	FORGED OR STAMPED STEEL BACKING PLATE SYSTEM	1
3	603-096-00	NUT, HEX, SS M8X1.25, WSHR W/TTH, 13 MM WDH	4
4	631-018-00	WASHER, FLAT, 8.4 ID, 16MM OD, SS	1

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	8mm Socket Head	1.6 N.m

METHOD:

6.1 Carefully remove the plastic protective film of the system power indicator (631-021-00) as shown in the figure and stick it to middle of the panel (631-018-00).

6.2 As shown in the figure, install the panel to the bottom leftmost position of the Interface panel of the cabinet operation surface, and fix it with 4 M5 hexagon nuts (603-096-00), The tool is an 8mm socket wrench with a torque value of 1.6 N·m.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	901-849-01	LC ADAPTER (EMI) 45DEG SCRW MT	1
2	630-893-00	FORGED OR STAMPED STEEL CAB I/O PANEL SIGNAL MISC	1
3	406-501-05	SCRW PAN HEAD TORX M2X.4PX5MM SS W/NYLOK	2

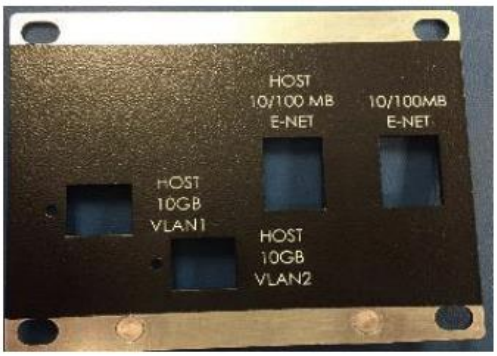
MACHINE:

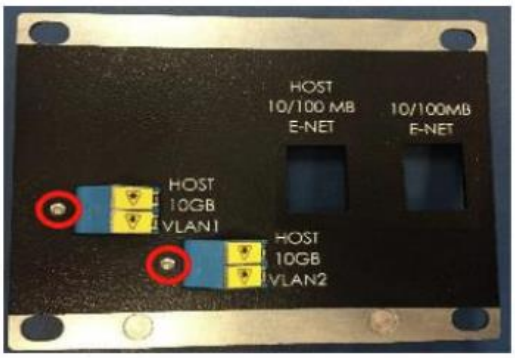
NO.	DESCRIPTION	PARAMETER
1	T6 Torx Bit	0.19 N.m

METHOD:

7.1 As shown in the figure, install 2 Ethernet adapters (901-849-01) to the "10GB E-Net" on the operation Interface panel (630-893-00) In the area, use 2 M2 Torx screws (406-501-05) to fix in front of the panel, the tool is a T6 Torx wrench, and the torque value is 0.19 N·m.

Note : Make sure the Adapters are properly flushed to panel





MAN: CERTIFIED OPERATOR, TECHNICIAN

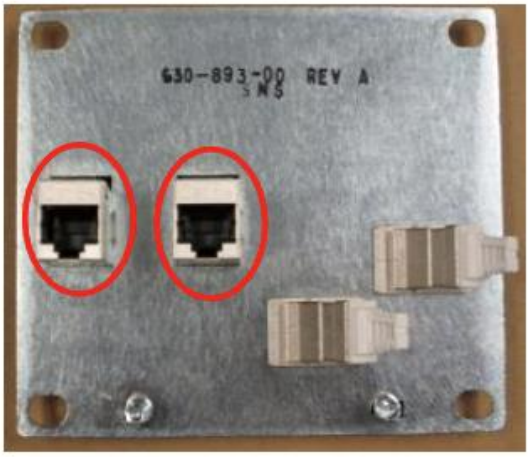
MATERIAL:

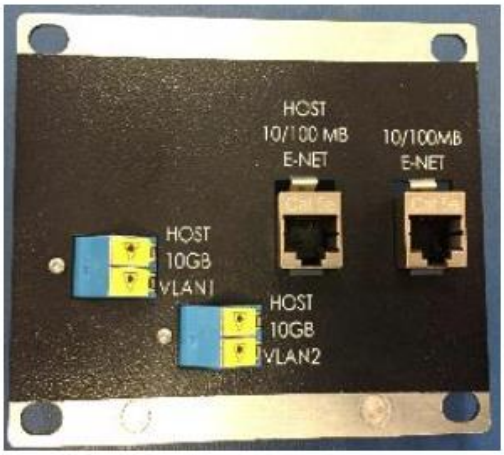
ITEM	P/N	DESCRIPTION	QTY
1	357-725-00	PANLMNT CAT5E RJ45 FEEDTHRU	2

MACHINE: N/A

METHOD:

7.2 Install two CAT5E connectors (357-725-00) in the remaining space of the panel as shown in the figure. Install from the back of the panel, install the connector diagonally .Then press the connector in.





MAN: CERTIFIED OPERATOR, TECHNICIAN

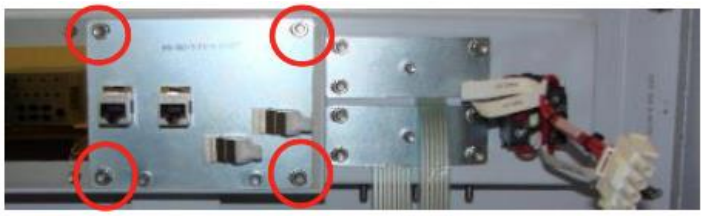
MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	603-096-00	NUT, HEX, SS M8X1.25, WSHR W/TTH, 13 MM WDH	8
2	634-733-00	CABLE CAB I/O PANEL EMO 24 V W/CONN	1

MACHINE: N/A

METHOD:

7.3 As shown in the figure, install an E-Net panel on the right side of the power panel on the cabinet. Use 4 5mm hexagon nuts (603- 096-00) to fix, Do not use tools to fix.



7.4 Install an AC power supply panel (634-733-00) on the right side of the E-Net panel of the cabinet as shown in the figure.
4 5mm hexagon nuts (603-096-00) Fixed, do not use tools to fix.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	634-736-00	CABLE CAB I/O PANEL 24 V W/CONN	1
2	603-096-00	NUT, HEX, SS M8X1.25, WSHR W/TTH, 13MM WDH	4
3	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 X 12 MM	1

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	PH2 Bit	3.3 N.m

METHOD:

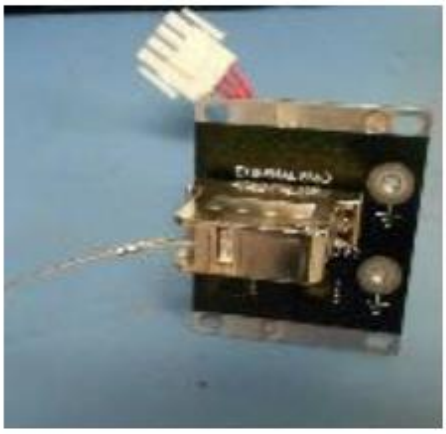
7.5 Install the external EMO panel

a. Take an external EMO panel (634-736-00).

b. As shown in the figure, install an EMO panel on the right side of the AC power panel on the cabinet.

Use 4 5mm hexagon nuts (603-096-00) to fix,

c. Fix the connection wire in the bottom hole as shown in the figure, and fix it with 1 M5 screw (472-426-12), the tool is a PH2 Phillips torque screwdriver, The torque value is 3.3 N·m



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MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

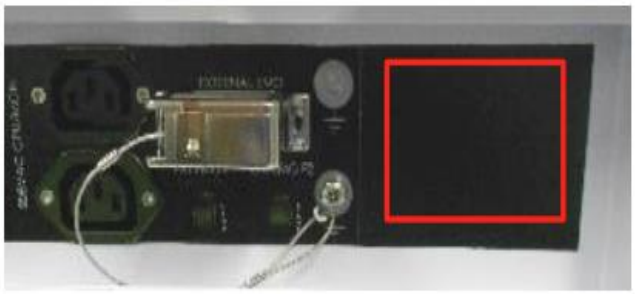
ITEM	P/N	DESCRIPTION	QTY
1	633-966-00	FORGED OR STAMPED STEEL CAB I/O PANEL SPARE MISC	1
2	603-096-00	NUT,HEX, SS M8X1.25, WSHR W/TTH, 13MM WDH	4

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	8mm Socket Head	3.3 N.m

METHOD:

7.5 As shown in the figure, install a black panel (633-966-00) on the right side of the external EMO panel on the cabinet, and pay attention to the two studs of the black panel at the below position. Use 4 5mm hexagon nuts (603-096-00) to fix do not use tools to fix.



7.6 Press the 4 panels to ensure that there is no gap between them. Use an 8mm socket wrench to fix the nut with a torque value of 3.3 N·m. A total of 16 nuts.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	801-541-02	SW CONTACT BLOCK, NO	1
2	801-515-02	SW CONTACT BLOCK 1F	1
3	634-438-00	CABLE EMO BUTTON CABLE W/ CONN	1

MACHINE:

NO.	DESCRIPTION	PARAMETER
1	Flat-Head Torque Screwdriver	0.7 N.m

METHOD:

5.4 Install the second connector (801-541-02) to the main connector (801-515-02) as shown in the figure.

5.5 Follow the steps below to install the wire (634-438-00) on the switch interface as shown in the figure.

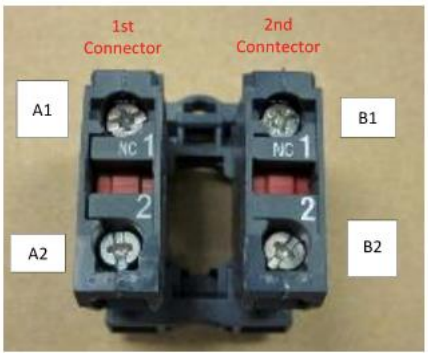
a. A1 is connected to terminal 1 of the 1st connector.

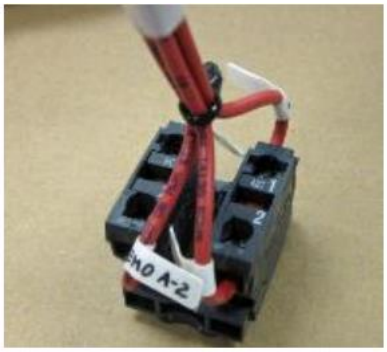
b. B1 is connected to terminal 2 of the 1st connector.

c. A2 is connected to terminal 1 of the 2nd connector.

d. B2 is connected to terminal 2 of the 2nd connector.

Use a flat-head torque screwdriver with a torque value of 0.7 N·m.





5.6 With the switch port facing upwards, install the EMO connector switch to the back position of the EMO button.

Gently push the button to ensure that the EMO switch can It is pressed in and rotated out normally without obstacles.

MAN: CERTIFIED OPERATOR, TECHNICIAN

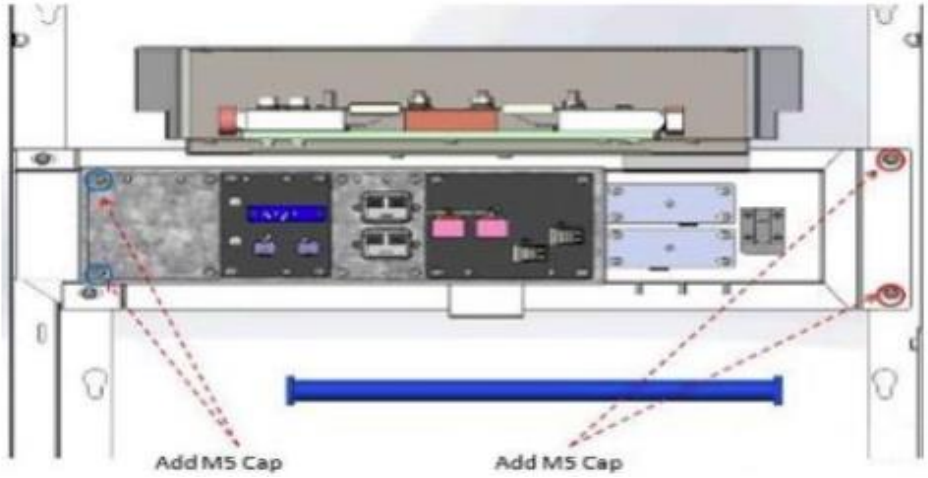
MATERIAL:

ITEM	P/N	DESCRIPTION	QTY
1	654-193-01	HD, NUT, HEX CAP M5, 7.9MM W, POLYETHYLENE, BLACK	4

MACHINE: N/A

METHOD:

7.7 Install 4 plastic protective covers (654-193-01) as shown in the figure.



7.8 Schematic diagram of panel assembly of the interface area of the final cabinet operation surface-front and back.